

Xinyuan Yan, Ph.D.

Postdoctoral Researcher · Computational Neuroscience & Psychiatry

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EDUCATION

Postdoctoral Researcher, Neurosurgery & Neuroscience April 2025 – Present
Baylor College of Medicine *Advisors: Benjamin Y. Hayden & Sameer A. Sheth*

Postdoctoral Researcher, Psychiatry & Neuroscience March 2022 – March 2025
University of Minnesota *Advisors: Alexander Herman & David Darrow*

Ph.D., Cognitive Neuroscience September 2016 – January 2022
IDG/McGovern Institute for Brain Research, State Key Laboratory of Cognitive Neuroscience and Learning
Beijing Normal University

B.Sc., Psychology September 2012 – June 2016
Northeast Normal University

FELLOWSHIPS & AWARDS

Travel Award, Computational and Systems Neuroscience (Cosyne), Montreal 2023

Trainee Professional Development Award, Society for Neuroscience 2022

MnDrive Neuromodulation Research Fellowship, University of Minnesota Medical School 2022

Fellowship, Computational and Cognitive Neuroscience (CCN) Summer School, CSHL Asia 2021

Fellowship, Kavli Summer Institute in Cognitive Neuroscience, UC Santa Barbara 2019

Distinguished Graduate Research Fellowship, State Key Laboratory of Cognitive Neuroscience and Learning 2018

National Graduate Research Fellowship, China 2018

PUBLICATIONS

† indicates co-last author. * indicates co-first author.

Concepts, language, single-neuron recordings, and hippocampus

1. **Yan, X.**, Chavez, A. G., Franch, ..., **Hayden, B. Y.†**, **Sheth, S. A.†** (2025). Shared neural geometries for bilingual semantic representations in human hippocampal neurons. *Cell*.
[https://www.cell.com/cell/fulltext/S0092-8674\(26\)00579-9](https://www.cell.com/cell/fulltext/S0092-8674(26)00579-9)
2. **Yan, X.**, Li, J.-A., Franch, M., Zhu, H., Cowan, R., Belanger, J., Chavez, A. G., Chericoni, A., Ismail, T., Katlowitz, K. A., Kolibius, L. D., Mickiewicz, E. A., Paulo, D., Bartoli, E., Hennig, J. A., Frączek, T. M., Provenza, N. R., Rahimpour, S., Shofty, B., . . . **Hayden, B. Y.†**, **Sheth, S. A.†** (2026). Polysemanticity in human hippocampal neurons. *bioRxiv*. <https://doi.org/10.64898/2026.05.02.722435>
3. Chericoni, A., Diao, C., **Yan, X.**, Ismail, T., Mickiewicz, E. A., Franch, M., . . . **Hayden, B. Y.†**, **Sheth, S. A.†** (2026). Neural geometry in the human hippocampus enables generalization across spatial position and gaze. *arXiv*. <https://arxiv.org/abs/2603.04747> *Neuron (in press)*.
4. Xiong, H. D.*, **Yan, X***, Li, J. A., Xue, J., Mattar, M. G., & Wilson, R. C. (2026). Thinking Under Uncertainty: Evidence Use and Information-Seeking in Language Models. *arXiv preprint arXiv:2607.26845*.

5. Chavez, A. G., **Yan, X.**, Franch, M., Mickiewicz, E. A., Baltazar, W., Belanger, J. L., Devara, D., . . . Hayden, B. Y. (2025). Hippocampus serves as a repository for spoken and heard word meanings during conversations. *bioRxiv*. <https://www.biorxiv.org/content/10.1101/2025.09.20.677504v3.abstract>
6. Ismail, T., Chavez, A. G., **Yan, X.**, Zhu, H., Franch, M., Belanger, J., ... & **Hayden, B. Y.** (2026). Estimation of neuronal tuning for word meaning from passively recorded naturalistic speech. *bioRxiv*, 2026-06. <https://www.biorxiv.org/content/10.64898/2026.06.23.733980v1.abstract>
7. Zhu, H., Chericoni, A., Ismail, T., Mickiewicz, E., Franch, M., Nigam, T., **Yan, X.**,... & **Hayden, B. Y.** (2026). A number simplex in the human medial temporal lobe. *bioRxiv*, 2026-06. <https://www.biorxiv.org/content/10.64898/2026.06.25.734462v1.abstract>

Beliefs about uncertainty and implications for affective disorders

Skill set: computational modeling, intracranial EEG

8. **Yan, X.**, Peterson, C. M., Schmidt, L. M., Koenig, S., Nixdorf, D. R., Herman, A., & Darrow, D. (2026). Impaired adaptive learning in chronic pain contributes to apathy. *eLife*.
9. **Yan, X.**, Ebitz, B. R., Darrow, D. P., & Herman, A. B. (2025). Nonlinear modulation of human exploration by distinct sources of uncertainty. *bioRxiv*, 2025-04.
10. **Yan, X.**, König, S. D., Ebitz, R. B., Hayden, B. Y., Darrow, D. P., & Herman, A. B. (2025). Dynamic prefrontal coupling coordinates adaptive decision-making. *Research Square*, rs-3.
11. **Yan, X.**, Ebitz, R. B., Grissom, N., Darrow, D. P., & Herman, A. B. (2025). Distinct computational mechanisms of uncertainty processing explain opposing exploratory behaviors in anxiety and apathy. ***Biological Psychiatry: Cognitive Neuroscience and Neuroimaging***.
12. Knep, E., **Yan, X.**, Chen, C. S., Jacob, S., Darrow, D. P., Ebitz, R. B., . . . Herman, A. B. (2025). Social aloofness is associated with non-social explore–exploit decisions. ***Communications Psychology***, 3(1), 106.
13. Cao, D.*, Cui, J.*, **Yan, X.***, Zhang, X., Dang, Y., Zhao, H., . . . Zhang, Y. (2025). Dynamic interactions in human amygdala subregions support hierarchical processing of fearful faces. *bioRxiv*, 2025-11.

Beliefs about social interactions

Skill set: computational modeling, fMRI

14. **Yan, X.** (2024). Too smart for one's own good? Evolution favors individuals with a low-level theory of mind in a complex network. *PsyArXiv*.
15. Liu, Y., Li, S.#, Lin, W.#, Li, W.#, **Yan, X.**, Wang, X., Pan, X., Rutledge, R. B., & Ma, Y. (2019). Amygdala activity tracks deviations from social preferences. ***Nature Neuroscience***, 22(4), 633–641. [*# co-second author*]
16. **Yan, X.**, Yong, X., Huang, W., & Ma, Y. (2018). Placebo treatment facilitates social trust and approach behavior. ***PNAS***, 115(22), 5732–5737.
17. Wang, D., **Yan, X.**, Li, M., & Ma, Y. (2017). Neural substrates underlying oxytocin effect: A quantitative meta-analysis of pharmaco-imaging studies. ***Social Cognitive and Affective Neuroscience***, 12(10), 1565–1573.

Beliefs about self and implications for major depressive disorder

Skill set: fMRI

18. **Yan, X.** (2024). Neural dynamics of self-referential processing and the insight for decoding self-concepts. ***Journal of Neuroscience***, 44(30). [*Commentary*]
19. Feng, C.†, **Yan, X.†**, Huang, W., Han, S., & Ma, Y. (2018). Neural representations of the multidimensional self in the cortical midline structures. ***NeuroImage***, 183, 291–299. [*† co-first author*]
20. Ma, Y., Li, S., Wang, C., Liu, Y., Li, W., **Yan, X.**, Chen, Q., & Han, S. (2016). Distinct oxytocin effects on belief updating in response to desirable and undesirable feedback. ***PNAS***, 113(33), 9256–9261.

EDITORIAL SERVICE

- **Guest Associate Editor**, *Frontiers in Human Neuroscience* — Research Topic: *Methods in Cognitive Neuroscience: 2023* (Oct 2023 – Dec 2023)

AD-HOC REVIEWER

PNAS · Nature Communications · Communications Biology · Translational Psychiatry · Brain and Behavior · Psychoradiology (Oxford Academic) · Self and Identity · Frontiers in Psychology · Frontiers in Psychiatry · Frontiers in Human Neuroscience · Frontiers in Endocrinology · Frontiers in Robotics and AI · BMC Neuroscience.

PROFESSIONAL MEMBERSHIPS

- Society for Neuroscience
- Society of Biological Psychiatry (Regular Member)

CONFERENCE PRESENTATIONS

Oral Presentations

- **Yan X.**, *Stable meaning, flexible codes: how the human hippocampus builds and reuses semantic geometry*. NYU Langone Health, August 5th, 2026, New York, USA.
- **Yan X.**, *Shared neural geometries for bilingual semantic representations in human hippocampal neurons*. NYU Psychology, August 6th, 2026, New York, USA.
- **Yan X.**, Koneig S, Ebitz B, Hayden, B, Darrow D, Herman A. *Dynamic prefrontal coupling coordinates adaptive decision-making*, Society for Neuroscience, November 15-19, 2025, San Diego, USA.
- **Yan, X.**, König, S., Ebitz, B., Hayden, B., Darrow, D., & Herman, A. (2024). *Deciphering neural mechanisms of explore–exploit decision-making in the prefrontal cortex*. Computational Psychiatry Conference, Minneapolis, USA.
- **Yan, X.**, Ebitz, B., . . . Darrow, D., & Herman, A. (2022). *The behavioral profile of the exploration–exploitation dilemma and its implications for anhedonia and anxiety*. Computational Psychiatry, New York City, USA.

Poster Presentations

- **Yan X.**, *Shared neural geometries for bilingual semantic and grammatical representations in human hippocampal neurons*. 9th annual conference on cognitive computational neuroscience. 2026, August 6th, New York, USA.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2023). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Computational and Systems Neuroscience (Cosyne), Montreal, Canada.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2022). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Society for Neuroscience Annual Meeting, San Diego, USA.